

Non-Ventilator Hospital-Acquired Pneumonia Prevention Through Oral Care

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Background

- Non-ventilator hospital-acquired pneumonia (NV-HAP) is one of the most common hospital-acquired infections
- Aspiration is the primary route for the acquisition of NV-HAP
- NV-HAP mortality rate is 15- 30% and is associated with a hospital length of stay of up to 15 days
- Requires ICU admission in up to 46% of non-ICU cases
- Increased readmission rates within 30 days is up to 20% of survivors
- NV-HAP acute care hospital treatment range from \$28,000-40,000

Purpose

Educate the nursing staff on a standardized, evidence-based oral care protocol and decrease the incidence of NV-HAP among acute stroke non-ventilated patients in a 12-bed Neuro Intensive Care Unit (ICU) of an acute care community hospital

Methods

Design

- Quality improvement project

Setting

- 12-bed Neuro intensive care unit (ICU) at an acute hospital in Southern California

Participants

- 45-nursing staff who work in the Neuro-intensive care unit (ICU)

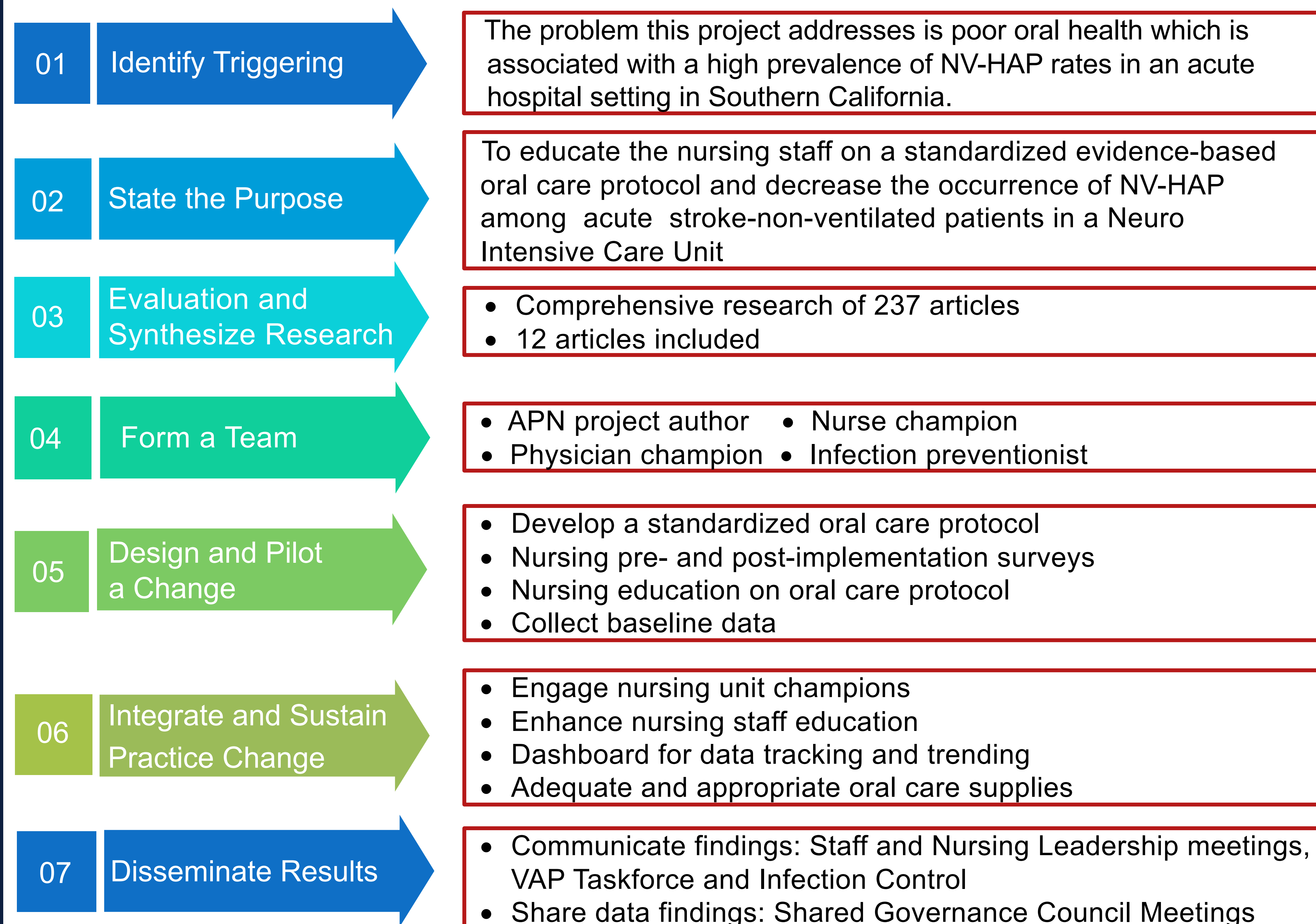
Interventions

- Develop and implement an evidence-based - oral care protocol for use on all non-mechanically ventilated acute stroke patients admitted to a Neuro ICU
- Educate the nursing team on the standardized oral care protocol
- Conduct a pilot project for two months

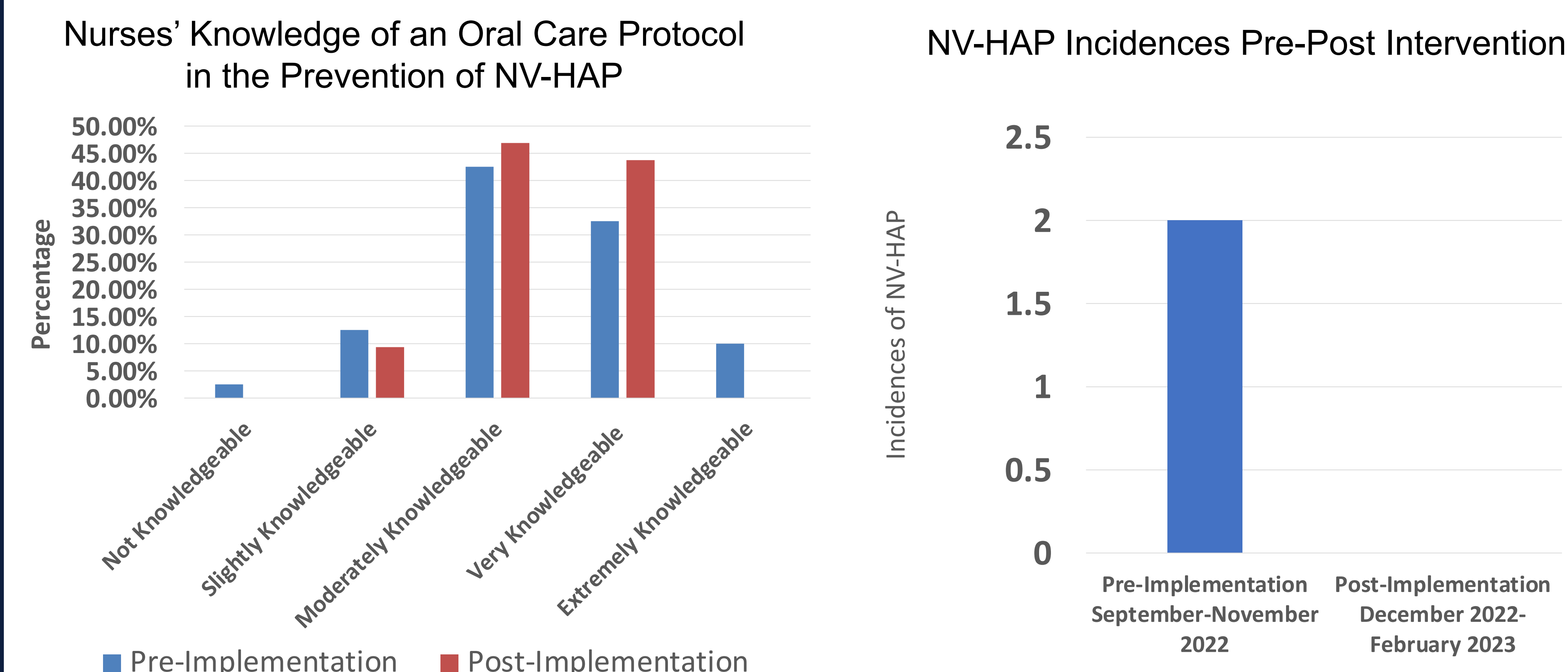
Measuring Instruments

- 12 questions pre- and post-implementation survey
- NV-HAP incidences

Theoretical Framework: The IOWA Model



Results



Discussion

- Education improved nurses' knowledge on the importance of an oral protocol to prevent NV-HAP
- Statistical significance was not reached: Small sample size & short duration of the pilot project.
- Barriers to implementing an oral care protocol:
 - Lack of time, clinical resources/supplies
 - Task prioritization
- Oral care kits placed next to the patient's room improved nursing adherence to the oral care protocol
- Nursing assistants performed oral care protocols for the patients

Conclusions & Recommendations

- Adherence to oral care protocol:
 - Strategic education
 - Nurse champion:
 - Weekly chart audits
 - Frequent rounding & communication with the nursing team, & nursing leadership
- Oral care protocol led to a clinical reduction of NV-HAP incidences.
- Improve reliability of findings and clinical outcomes:
 - Larger sample size of nurses representing all hospital units
 - Longer duration of the pilot project
- Develop surveillance programs and prevention guidelines

Implications for Practice

- Nursing leadership and support: provide adequate clinical resources and supplies to the hospital units
- Provide adequate knowledge and training to the nursing staff
- Nursing champions to represent various clinical settings

References



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